

Determining the Mechanisms of Temporal Context in Autobiographical Memory with Intracranial EEG

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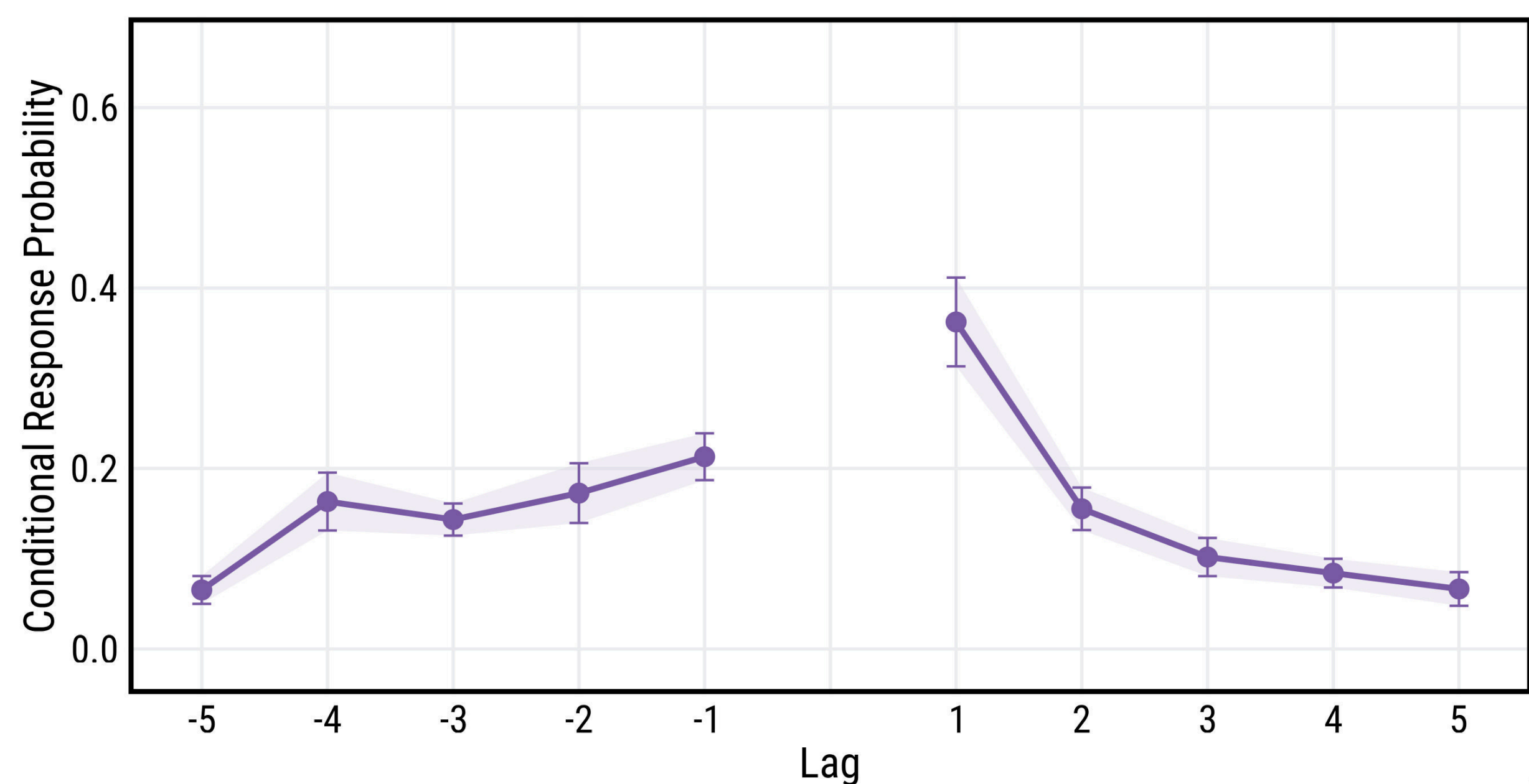
Background

- Recalling life memories not only depends on recovering the content of an event but also its context. Such context acts as an anchor for episodic detail (**temporal contiguity effect**).^{1,2}
- The **Context Maintenance & Retrieval (CMR)** framework is a formal model of this effect⁴ and predicts that context pushes memory recall **towards subsequent events (forward asymmetry)**.⁵
- CMR has robust support but much of it is limited to list-learning paradigms.
- Unlike lists, real-world memories are self-generated, multidimensional, and embedded in a continuous experience of the world.⁷
- As such, there is a **missing link between autobiographical memory and computational models of memory**.⁶

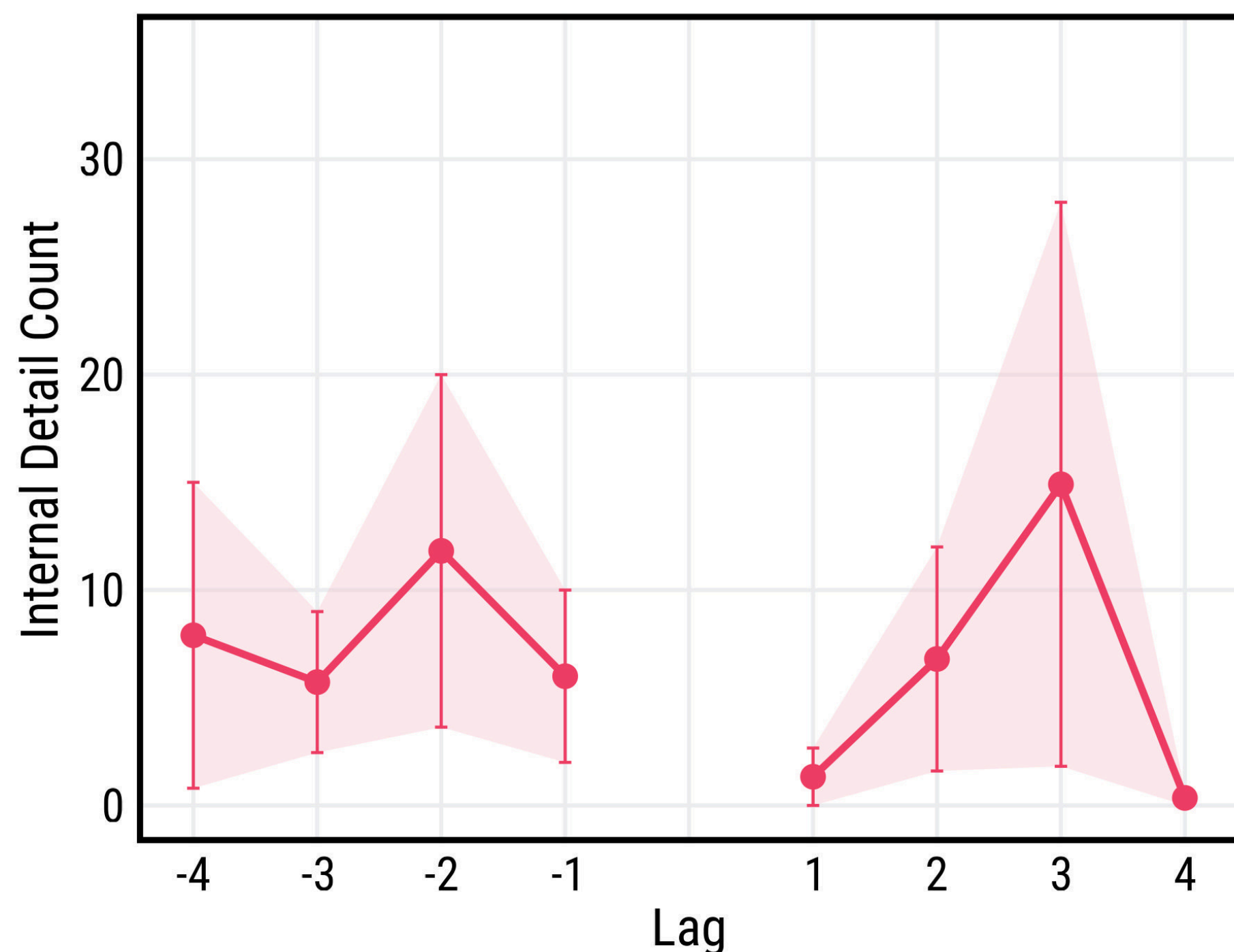
Methods

- We hope to recruit $N = 25$ healthy controls and $N = 24$ RNS participants.
- Two methodological advances: **(1) HippoCamera** is a smartphone application that **guides users to capture audiovisual cues that allow them to later re-experience real-world events** by engaging the hippocampus.⁸
- (2) The NeuroPace RNS system** provides **chronic intracranial recordings from the human hippocampus**.⁹
- Behavioural: conditional response probability (CRP)**¹⁰, **serial position curves**¹¹, and **internal details from the Autobiographical Interview**.¹²
- Neural: dynamic causal modelling (DCM)**¹³, **subspace representational analysis**¹⁴, and **principal components analysis (PCA)**.¹⁵

Preliminary Results



- So far, participants ($n = 13$) are showing some expected **forward asymmetry**, with pronounced CRP at lag +1.
- No statistical analyses to date due to small sample size.
- However, we are noticing **backward asymmetry (prob = 0.141) versus forward asymmetry (prob = 0.118)**, where participants' narratives tend to transition towards earlier events.

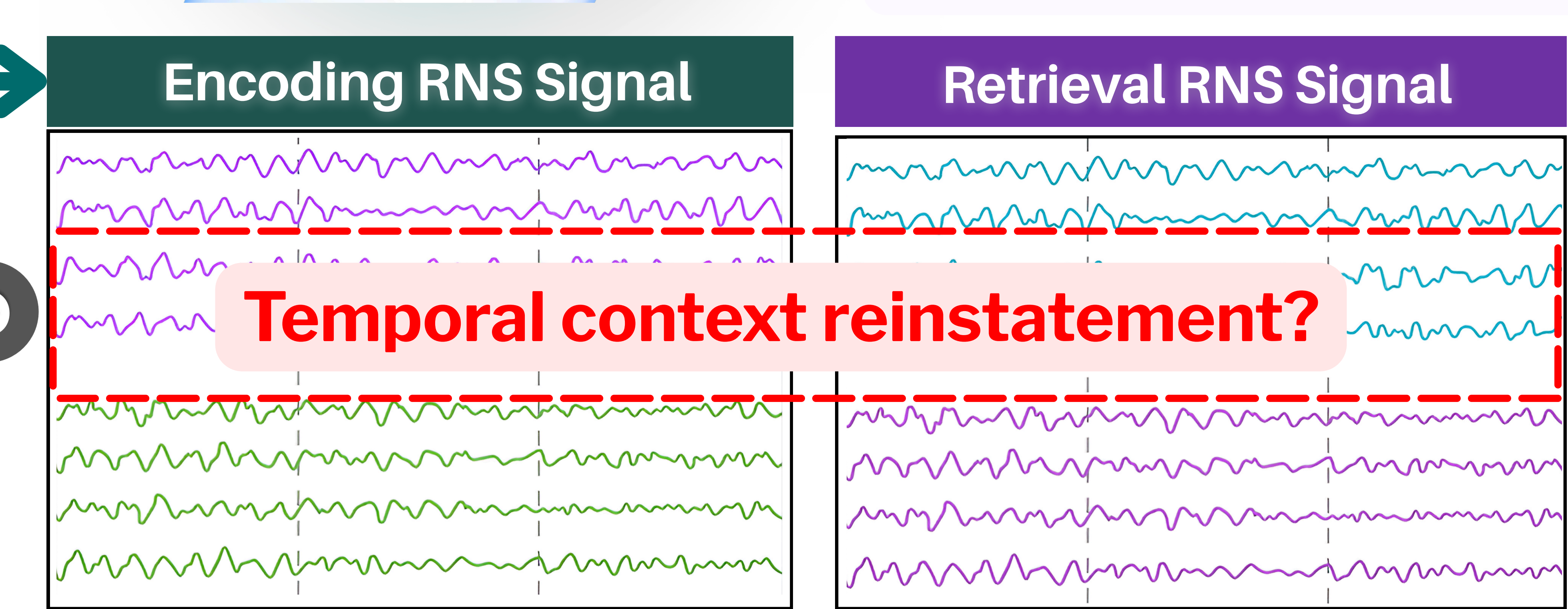
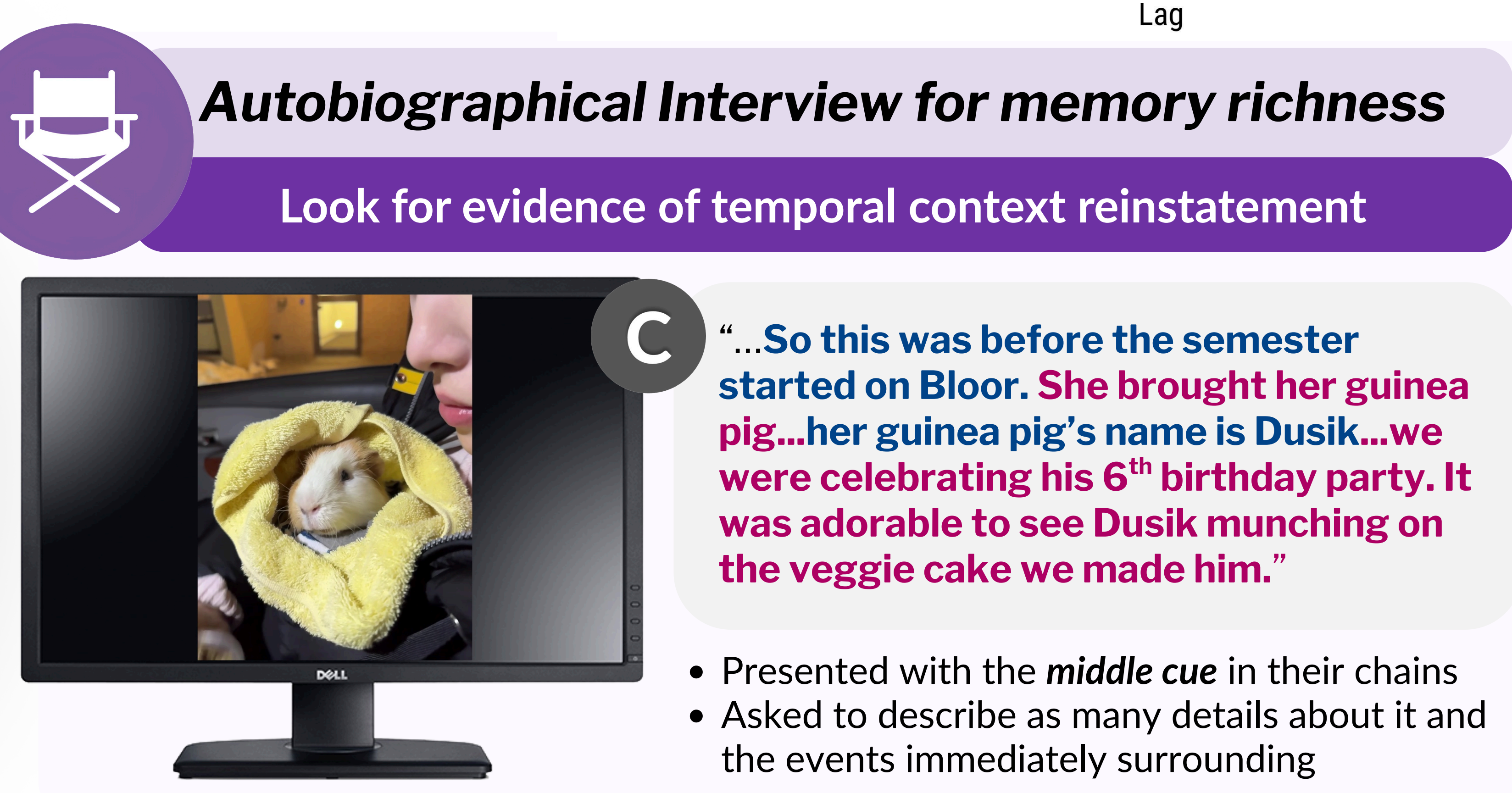
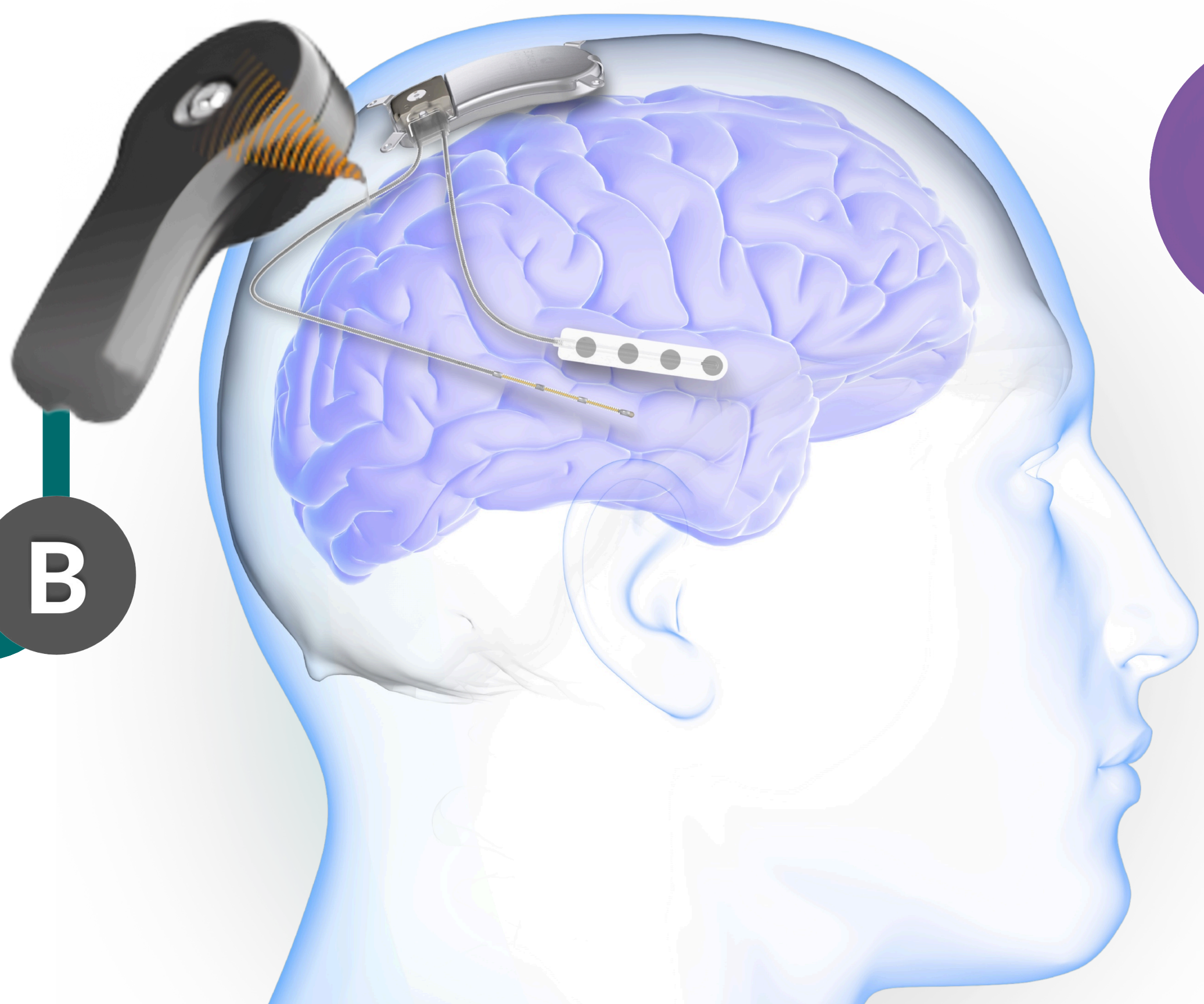


- Similar to our lag CRP data, participants ($n = 2$) appear to be exhibiting **some forward asymmetry for episodic richness by lag**.
- Again, there is some **backward asymmetry compared to forward asymmetry for the depth of details provided**.

Future Directions

- Encoding-similarity analysis in epilepsy patients
- Left: Application of DCM to identify consistent hippocampal longitudinal effective connectivity (warm colours indicate anterior → posterior propagation of a travelling wave).

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(A) **EVENT CHAINS CREATION:** participants record 5-9 cues daily (24s videos with 8s audios). (B) **ENCODING RNS SIGNAL:** participants swipe magnetic wand when recording. (C) **RETRIEVAL RNS SIGNAL:** autobiographical interview for episodic richness. (D) **ENCODING-SIMILARITY ANALYSIS:** During the autobiographical interview in RNS patients, we will relate the iEEG signals picked up at encoding to the iEEG signals picked up at retrieval.